

CORE EQUATIONS (0.1–10.0 Scale)

HICE (Individual)	$H = I \times C \times E$	$IQ \text{ (Born With)} \times CQ \text{ (Created)} \times EQ \text{ (Developed)}$
Signal Formula	$S = (A \times P) / \beta$	$Amplitude \times Persistence / Physical Drag$
Seesaw Equation	$AD^2 = 16$	$As \text{ capability } (A) \text{ rises, resistance } (D^2) \text{ must rise faster}$
ISI (Civilisational)	$ISI = ISI_1 \otimes ISI_2 \otimes ISI_3$	$Sustainability \times Survival \times Significance$
Surplus Equilibrium	$A_surplus \times P_gov / \beta_harm \rightarrow stable$	$Surplus domesticated when band is stable$
Dominion Condition	H must remain senior partner	$Holistic \text{ intelligence} > synthetic \text{ intelligence}$

RESISTANCE (Aggregate Scale)

Physical Drag β	Single aggregate value (0.1–10.0)	$Composite \text{ of } 4Cs: \text{ Conflict, Climate, Contagion, Cost}$
Ethical Drag λ	Single aggregate value (0.1–10.0)	$Composite \text{ of: Doctrine, Grace, Interpretation}$
Additive Resistance	$R_add = \beta + \lambda$	$Baseline \text{ load} \text{ — how much total resistance}$
Multiplicative Resistance	$R_mult = \beta \times \lambda$	$Stress \text{ test} \text{ — compounding when both high}$
Credit (Additive)	$C_add = (A \times P) / R_add$	$Governance \text{ capacity under additive load}$
Credit (Multiplicative)	$C_mult = (A \times P) / R_mult$	$Governance \text{ capacity under compounding load}$
Sensitivity Ratio	$\rho = C_add / C_mult$	$\rho < 2 \text{ GREEN} \mid 2 \leq \rho < 5 \text{ AMBER} \mid \rho \geq 5 \text{ RED}$

SCALE GUIDE

A (Amplitude)	0.1–10.0	1-3: Local 4-6: Regional 7-8: Continental 9-10: Global
P (Persistence)	0.1–10.0	1-3: Decades 4-6: Centuries 7-8: Millennia 9-10: Permanent
β (Physical Drag)	0.1–10.0	1-3: Low drag 4-6: Moderate 7-8: High 9-10: Catastrophic
λ (Ethical Drag)	0.1–10.0	1-3: Strong governance 4-6: Moderate 7-8: Weak 9-10: Absent

TRAFFIC LIGHT THRESHOLDS

GREEN	$\rho < 2.0$	$System \text{ robust} \text{ — additive and multiplicative loads similar}$
AMBER	$2.0 \leq \rho < 5.0$	$System \text{ sensitive} \text{ — compounding drag significantly amplifies risk}$
RED	$\rho \geq 5.0$	$System \text{ fragile} \text{ — multiplicative collapse likely under stress}$

FRACTAL CONNECTOR (FC13)

FC13 Definition	Intelligence layer binding 12 relays	$Sun \text{ Tzu Ch.13 (Use of Spies)} = invisible \text{ web}$
ICU Nexus	$0 + 1 = System \text{ Node}$	$0 \text{ (The Field/civilisation)} + 1 \text{ (The Individual)} = engagement$
Maslow-Dearden Dock	Individual $\leftrightarrow$ Civilisational	$Maslow \text{ apex} = Cone \text{ apex} = system-ready \text{ (Scholar 8)}$
Per Arya Ad Astra	Why Thirteen	$13 \text{ cards per suit. } 12 \text{ relays} + FC13 \text{ completes the deal.}$

INTEGRATION-LEVEL MULTIPLIER (v5)

Level Multiplier	$M(L) = 1 + \ln(L)$	Amplifies the Signal by integration depth L (1–8). $\ln(1)=0 \rightarrow L1 \text{ baseline un-amplified; diminishing returns thereafter (Weber–Fechner / Shannon).}$
Apply to Signal	$S_scaled = S_raw \times M(L)$	$S_raw = (A \times P) / \beta$. Multiplier applies BEFORE the saturation cap (see Integration Levels, Section G).
M(L) values	L1=1.00 L2=1.69 L3=2.10 L4=2.39 L5=2.61 L6=2.79 L7=2.95 L8=3.08	Largest jump L1→L2 (+69%); smallest L7→L8 (+4%). Carries the L7/L8 separation since Web breadth saturates at 5.

PULSE RAW SIGNAL (Mode 1 — Signal / Discovery)

PULSE raw signal	$PULSE = ((A \times P) / \beta) / 25$	Normalises the raw Signal to a 0–1-ish discovery band by dividing by 25. Divisor 25 = max plausible $(A \times P) / \beta$ on the quick-scan estimate scale, mapping a strong scan to ~1.0.
Divisor basis	$25 = 10 \times 10 / 4 \text{ (approx.)}$	A,P quick-scan capped near 10; β floor near 0.4 on the normalised scan $\rightarrow$ ceiling ~25, so /25 keeps PULSE in the colour-band range for iGO display.
PULSE $\rightarrow$ colour band	Deep Red $\leq 0.2 < Red/Orange \leq 0.4 < Amber \leq 0.6 < Green \leq 0.8 < Gold$	Vibrational bands: Discordant / Disruptive / Harmonic / Resonant / Coherent. PULSE shows only the band (no detailed panel).

THE THREE 4Cs SETS (disambiguation — v5)

Set 1 — Signal Formula β	Conflict Contagion Climate Cost	The FOUR DRAG forces composited into physical drag β in $S=(A \times P) / \beta$. These are the BEACON 4Cs stress levers (B_1 – B_4).
Set 2 — Four Horsemen (narrative)	Conflagration Consumption Conflict Climate	The narrative collapse-vector set used in the Master Database 'Collapse Vector' column. Distinct from the β set: this is storytelling/diagnosis of HOW a relay fails.
Set 3 — Relay 12 Human Capability	Consciousness Cognition Communication Collaboration	The FOUR human-capability C's of Relay 12 (Human Nodes). These are CONSTRUCTIVE capacities, not resistance forces — do not confuse with the β drag set.
Why three sets	Same initial letter, different roles	Set 1 = resistance (degrades signal). Set 2 = failure narrative (collapse vectors). Set 3 = human capability (Relay 12). Kept explicit to prevent conflation in decks/app.

v5 ADDITION SOURCES

Sources	COUNTERFORCE v67–68; CSE Book 1 Ch.3 (Dearden Field); Weber–Fechner; Shannon (1948)	M(L), PULSE divisor, and the three 4Cs sets are defined in the iAAi framework; logarithmic basis per Weber–Fechner & Shannon.
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PLATO QUOTE (Foundation of Scale)

"The small-scale IS the foundation. Without it, the larger scales collapse." — Plato

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DOT-INTERFACE PROTOCOL (DIP)

DIP Event	$I \rightarrow HICE(latent \text{ heat}) \rightarrow "." \rightarrow C$	Human signal ($I=1$) + HICE tensor = Phase Change to Creation
DIP Boundary	$[1,1,1] \rightarrow [5,12,5]$	Fibonacci-derived growth bounds — exceeding triggers Safety-Throttling (4
SES Limiter	If HICE = 0, State-Change Aborted	Null-State Error — prevents Flash Boiler (potential→kinetic without human

TESLA-DEARDEN 3-6-9 CIVILISATIONAL CLOCK

0 (Zero Hour)	Creation = The Void/Potential	The Origin (Theological First Cause) — the Garden exists
3 (Signal)	WHAT = The Seed	Defining the Frequency of the intervention
6 (Coupling)	HOW = The Vibration	Docking/Synchronization via DIP ". " Break — eliminates Babel/Whispers
9 (Purpose)	WHY = The Field	Aligning Function with Purpose — SES verification
12 (Record)	RECORD = The Manifestation	The Built Reality (The Logos/Cursive) — permanent civilisational record

DEARDEN TENSOR LATTICE

Lattice Order	$[5, 12, 5] = 300 \text{ nodes}$	Web ($W=1-5$) $\times$ Relay ($R=1-12$) $\times$ Integration Level ($L=1-5$)
Node Address	$[W, R, L]$	Each node independently addressable — human navigates with ICE tensor
Determinant Property	$\det(ICE) = I \times C \times E = H$	If any component = 0, volume collapses ($H=0$) — zero-volume proof for AI

500-GENERATION COMPRESSION

Generational Count	$12,000 \text{ yr} / 25 \text{ yr/gen} \approx 500 \text{ generations}$	Entire civilisational memory compressed into ~500 human lifetimes
Compression Ratio	$4.5B \text{ yr Earth} / 12,000 \text{ yr civ} = 375,000$	Fragility Multiplier ($1/\beta$) — civilisation is 375,000× thinner than its substrate